

Partial Differential Equation (PDE)

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An equation involving the partial derivatives of a function of more than one variable is called partial differential equation (PDE). The general second order linear PDE in two independent variables x, t is

$$a(x, t) \frac{\partial^2 u}{\partial x^2} + b(x, t) \frac{\partial^2 u}{\partial x \partial t} + c(x, t) \frac{\partial^2 u}{\partial t^2} + d(x, t) \frac{\partial u}{\partial x} + e(x, t) \frac{\partial u}{\partial t} + f(x, t)u = g(x, t).$$

We will study the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

and the heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}.$$

0.1 Wave Equations

One-dimensional wave equation has the form:

$$u_{tt} = c^2 u_{xx}. \quad (1)$$

If $u = u(x, t)$ denotes vertical displacement of the vibration of a string then u satisfies the wave equation. The string can be finite and infinite. The string can be fixed at both ends, or just at one end. Each will leads us to different boundary conditions. The initial conditions for the one-dimensional wave equation are

$$u(x, 0) = f(x), u_t(x, 0) = g(x).$$

For the finite string the boundary conditions are

$$u(0, t) = A(t), u(L, t) = B(t).$$

Dirichlet Problem and Separation of Variables

The method of separating the variables is a very convenient way to obtain solutions for IBVP on finite interval. If we tie the string at both ends we can have the following boundary conditions:

$$u(0, t) = 0, \quad u(L, t) = 0.$$

Now we solve following Dirichlet problem using the method of separating the variables.

Problem. *Solve*

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < L, t > 0, \quad (2)$$

$$u(0, t) = 0, u(L, t) = 0, \quad t > 0, \quad (3)$$

$$u(x, 0) = f(x), u_t(x, 0) = g(x) \quad 0 < x < L. \quad (4)$$

Solution. *Let us assume the solution $u(x, t)$ of (2) as a product of a function that depends only on the variable t and of a function that depends only on the variable x i.e., $u(x, t) = X(x)T(t)$. We substitute u in the equation $u_{tt} = c^2 u_{xx}$, to obtain:*

$$X(x)\ddot{T}(t) = c^2 \ddot{X}(x)T(t), \quad (5)$$

Eqn (5) can be written as

$$\frac{\ddot{T}(t)}{c^2 T(t)} = \frac{\ddot{X}(x)}{X(x)}.$$

Observe that the right hand side is function of x alone while the left hand side is a function of t alone. Hence each of them must be constant and equal to, say, α . Therefore

$$\frac{\ddot{T}(t)}{c^2 T(t)} = \frac{\ddot{X}(x)}{X(x)} = \alpha.$$

Now we solve each ordinary differential equation

$$\ddot{X} - \alpha X = 0, \quad (6)$$

$$\ddot{T} - c^2 \alpha T = 0, \quad (7)$$

separately. From (4) we have

$$\begin{aligned} u(0, t) &= X(0)T(t) = 0 \quad \forall \quad t \geq 0, \\ u(L, t) &= X(L)T(t) = 0 \quad \forall \quad t \geq 0. \end{aligned}$$

Since $T(t) \neq 0$, we get

$$X(0) = X(L) = 0. \quad (8)$$

We have the following three cases depending on α , (I) $\alpha > 0$ (II) $\alpha = 0$. (III) $\alpha < 0$

Case I Assume that $\alpha > 0$, i.e., $\alpha = \lambda^2$. In view of (6) and (7) we have $\frac{\ddot{X}(x)}{X(x)} = \lambda^2$, and $\frac{\ddot{T}(t)}{c^2 T(t)} = \lambda^2$. Thus X and T are given by $X(x) = c_1 e^{\lambda x} + c_2 e^{-\lambda x}$, and $T(t) = d_1 e^{\lambda ct} + d_2 e^{-\lambda ct}$. Thus $u(x, t) = (d_1 e^{\lambda ct} + d_2 e^{-\lambda ct})(c_1 e^{\lambda x} + c_2 e^{-\lambda x})$. Using (8) we get $c_1 + c_2 = c_1 e^{\lambda L} + c_2 e^{-\lambda L} = 0$. Solving for c_1 and c_2 we have $c_1 = c_2 = 0$. Hence $u = 0$. Since we are looking for non zero solution, $\alpha > 0$ is not a suitable choice.

Case II Assume that $\alpha = 0$. The equations (6) and (7) become $\ddot{X}(x) = \ddot{T}(t) = 0$. Therefore, $X(x) = c_1 x + c_2$, and $T(t) = d_1 t + d_2$, in turn $u(x, t) = (d_1 t + d_2)(c_1 x + c_2)$. using (8) we get $c_1 = c_2 = 0$. Thus $u = 0$. Since we are looking for non zero solution, $\alpha = 0$ is not a suitable choice.

Case III Assume that $\alpha < 0$, i.e., $\alpha = -\lambda^2$. In this case (6) becomes $\ddot{X}(x) = -\lambda^2 X(x)$. Solving for X we get $X(x) = c_1 \sin(\lambda x) + c_2 \cos(\lambda x)$. In view of (7), we get $\ddot{T}(t) = -\lambda^2 c^2 T(t)$ and $T(t) = d_1 \sin(\lambda ct) + d_2 \cos(\lambda ct)$. Thus

$$u(x, t) = (d_1 \sin(\lambda ct) + d_2 \cos(\lambda ct))(c_1 \sin(\lambda x) + c_2 \cos(\lambda x)). \quad (9)$$

Using the boundary condition (8), we get

$$\begin{aligned} (c_1 \sin(0) + c_2 \cos(0)) &= 0 \\ (c_1 \sin(\lambda L) + c_2 \cos(\lambda L)) &= 0, \end{aligned}$$

which implies

$$\begin{aligned} c_2 &= 0 \\ c_1 \sin(\lambda L) &= 0. \end{aligned}$$

Therefore, nonzero eigenvalues are $\lambda = \frac{\pi n}{L}$. Thus, the equation (9) reduces to

$$u(x, t) = \sum_{n=1}^{\infty} \left[d_{1n} \sin\left(\frac{\pi n}{L} ct\right) + d_{2n} \cos\left(\frac{\pi n}{L} ct\right) \right] c_n \sin\left(\frac{\pi n}{L} x\right). \quad (10)$$

Equation (10) can be written as

$$u(x, t) = \sum_{n=1}^{\infty} \left[\frac{L}{n\pi c} A_n \sin\left(\frac{n\pi ct}{L}\right) + B_n \cos\left(\frac{n\pi ct}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right). \quad (11)$$

Making use of initial condition (4) in (10), we get

$$\begin{aligned} u(x, 0) &= f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right), \\ u_t(x, 0) &= g(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right). \end{aligned}$$

Thus the coefficients A_n and B_n are

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right), \quad A_n = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right). \quad (12)$$

Hence (11) with (12) represents the solution of initial boundary value problem (2)-(4).

Particular Case I: If initial velocity $u_t(x, 0) = g(x) = 0$, then $A_n = 0$. Thus, in this case the solution (11) reduces to

$$u(x, t) = \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi ct}{L}\right) \sin\left(\frac{n\pi x}{L}\right). \quad (13)$$

Particular Case II: If initial displacement $u(x, 0) = f(x) = 0$, then $B_n = 0$. Thus, in this case the solution (11) reduces to

$$u(x, t) = \sum_{n=1}^{\infty} \frac{L}{n\pi c} A_n \sin\left(\frac{n\pi ct}{L}\right) \sin\left(\frac{n\pi x}{L}\right).$$

Exercise. Solve the IBVP:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < \pi, \quad t > 0$$

$$u(0, t) = u(\pi, t) = 0, \quad t \geq 0,$$

$$u(x, 0) = 5 \sin x + 12 \sin 2x + 6 \sin 3x,$$

$$\frac{\partial u}{\partial t}(x, 0) = 0, \quad 0 \leq x \leq \pi.$$

Solution. Since $g = 0$, $A_n = 0$. Comparing the coefficients B_n , we get

$$B_1 = 5, \quad B_2 = 12, \quad B_3 = 6,$$

and all the others are zero. Hence the solution is

$$u(x, t) = 5 \sin x \cos t + 12 \sin 2x \cos 2t + 6 \sin 3x \cos 3t.$$

Exercise. Solve

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0,$$

$$u(0, t) = u(\pi, t) = 0, \quad t \geq 0,$$

$$u(x, 0) = a \sin^3 x, \quad u_t(x, 0) = 0, \quad 0 \leq x \leq \pi.$$

Solution. Since $u_t(x, 0) = g(x) = 0$, $A_n = 0$. Now

$$a \sin^3 x = \frac{3a}{4} \sin x - \frac{a}{4} \sin 3x = \sum_{n=1}^{\infty} B_n \sin(nx/2).$$

Comparing the coefficients, we get $B_2 = 3a/4$, $B_6 = -a/4$, and $B_n = 0$, for $n \neq 2$ or $n \neq 6$. In view of (13), we get the required solution as

$$u(x, t) = \frac{3a}{4} \sin x \cos ct - \frac{a}{4} \sin 3x \cos 3ct.$$

Exercise. Solve

$$\begin{aligned}u_{tt} &= c^2 u_{xx}, & 0 < x < \pi, & \quad t > 0, \\u(0, t) &= u(\pi, t) = 0, & t &\geq 0, \\u(x, 0) &= b(\sin x - \sin(2x)), & u_t(x, 0) &= 0, \quad 0 \leq x \leq \pi.\end{aligned}$$

Solution. In this case $A_n = 0$, $B_1 = b$, $B_2 = -b$, and $B_n = 0$ for all $n \geq 3$. Thus the solution is given by

$$u(x, t) = b(\sin x \cos ct - \sin 2x \cos 2ct).$$

Problem. A string of length π is held fixed at both endpoints. Its initial position is $f(x) = \sin(x)$ and its initial velocity is $g(x) = \cos x$. Assuming that $c = 1$, find the position of the string $u(x, t)$ for every $x \in [0, \pi]$ and for every $t > 0$. Find an approximate value for $u(x, t)$ by adding several terms of the series.

Problem. Solve

$$\begin{aligned}u_{tt} &= u_{xx}, & x &\in (0, \pi), \quad t > 0, \\u(0, t) &= 0, \quad u_x(\pi, t) = 0, & t &> 0, \\u(x, 0) &= \sin(x), \quad u_t(x, 0) = 0, & x &\in [0, \pi].\end{aligned}$$

Uniqueness of the solution to the wave equation

Theorem. Let $u_1(x, t)$ and $u_2(x, t)$ be two solutions of the problem:

$$\begin{aligned}u_{tt} &= c^2 u_{xx}, & 0 < x < L, t > 0, \\u(0, t) &= A(t), \quad u(L, t) = B(t), & t &> 0, \\u(x, 0) &= f(x), \quad u_t(x, 0) = g(x), & 0 < x < L,\end{aligned}$$

where A , B , f , g are piece wise continuous functions. Then $u_1(x, t) = u_2(x, t)$ for all points in the domain.

Proof. Let $v(x, t) = u_1(x, t) - u_2(x, t)$, then v satisfies the following IBVP:

$$\begin{aligned} v_{tt} &= c^2 v_{xx}, & 0 < x < L, t > 0, \\ v(0, t) &= 0, \quad v(L, t) = 0, & t > 0, \\ v(x, 0) &= 0, \quad v_t(x, 0) = 0, & 0 < x < L. \end{aligned}$$

It is enough to prove that $v(x, t) = 0 \quad \forall x, t$.

We define:

$$E(t) = \int_0^L [c^2 v_x(x, t)^2 + v_t(x, t)^2] dx.$$

We will prove that $E(t) = 0$ first. On differentiating E with respect to t , we obtain

$$\begin{aligned} \dot{E}(t) &= \int_0^L [c^2 2v_x v_{xt} + 2v_t v_{tt}] dx \\ &= 2c^2 \int_0^L [v_x v_{xt} + v_t v_{xx}] dx \\ &= 2c^2 \int_0^L \frac{\delta}{\delta x} (v_x v_t) dx = 2c^2 [v_x(x, t) v_t(x, t)]_0^L \\ &= 2c^2 (v_x(L, t) v_t(L, t) - v_x(0, t) v_t(0, t)) = 0. \end{aligned}$$

Since $\dot{E}(t) = 0$, $E(t)$ is constant, and as $E(0) = 0$, we conclude that $E(t) = 0$.

Then $v_t(x, t) = 0$, and $v(x, t) = v(x, t) - v(x, 0) = \int_0^L v_t(x, t) dt = 0$. Hence $u_1(x, t) = u_2(x, t)$. □

D'Alembert's Solution of one dimensional wave equation

D'Alembert's Solution describes the vibration of infinite string. When we have an infinite string, with no boundary, then we have the following Cauchy Problem

$$u_{tt} = c^2 u_{xx}, \quad -\infty < x < \infty, \quad t > 0, \tag{14}$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x), \quad -\infty < x < \infty. \tag{15}$$

Solution

The wave equation can be decomposed as follows:

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2}\right) u(x, t) = \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}\right) u(x, t) = 0.$$

Let: $v(x, t) = \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}\right) u(x, t)$, then the wave equation becomes $\left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x}\right) v(x, t) = 0$.

Now $v(x, t) = F(x + ct)$, satisfies the PDE: $\left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x}\right) v(x, t) = 0$. Thus, wave equation reduces to

$$\left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}\right) u(x, t) = F(x + ct). \quad (16)$$

We introduce new coordinate $\xi = x - ct$, $\eta = x + ct$, and let $U(\xi, \eta) = u(x, t)$. Now

$$\begin{aligned} u_t + cu_x &= U_\xi \xi_t + U_\eta \eta_t + c(U_\xi \xi_x + U_\eta \eta_x) \\ &= -cU_\xi + cU_\eta + cU_\xi + cU_\eta \\ &= 2cU_\eta. \end{aligned}$$

Thus, the wave equation (16) becomes

$$2cU_\eta(\xi, \eta) = F(\eta). \quad (17)$$

Solving (17), we have

$$U(\xi, \eta) = \int \frac{1}{2c} F(\eta) d\eta + G(\xi) = H(\eta) + G(\xi). \quad (18)$$

Hence

$$u(x, t) = H(x + ct) + G(x - ct). \quad (19)$$

Using the initial conditions (15) in (19), we get

$$\begin{aligned} f(x) &= u(x, 0) = H(x) + G(x) \\ g(x) &= u_t(x, 0) = \dot{H}(x)c - \dot{G}(x)c, \end{aligned}$$

which implies

$$\begin{aligned} H(x) + G(x) &= f(x) \\ H(x) - G(x) &= \frac{1}{c} \int_0^x g(s) ds + C. \end{aligned}$$

Thus

$$\begin{aligned} H(x) &= \frac{1}{2} \left[f(x) + \frac{1}{c} \int_0^x g(s) ds + C \right] \\ G(x) &= \frac{1}{2} \left[f(x) - \frac{1}{c} \int_0^x g(s) ds - C \right] \\ &= \frac{1}{2} \left[f(x) + \frac{1}{c} \int_x^0 g(s) ds - C \right]. \end{aligned}$$

Substituting G and H in (19), we get

$$\begin{aligned} u(x, t) &= H(x + ct) + G(x - ct) \\ &= \frac{1}{2} \left[f(x + ct) + \frac{1}{c} \int_0^{x+ct} g(s) ds + C \right] + \frac{1}{2} \left[f(x - ct) + \frac{1}{c} \int_{x-ct}^0 g(s) ds - C \right]. \end{aligned}$$

Thus, the solution of the wave equation for the infinite string is:

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds. \quad (20)$$

which is called D'Alembert's Solution of wave equation.

Exercise. Solve the Cauchy problem

$$\begin{aligned} u_{tt} &= 4u_{xx}, \quad -\infty < x < \infty, \quad t > 0, \\ u(x, 0) &= ae^{-x^2}, \quad u_t(x, 0) = 0, \quad -\infty < x < \infty. \end{aligned}$$

Solution. The D'Alembert's Solution (20) takes the form

$$u(x, t) = \frac{a}{2} \left[e^{-(x+2t)^2} + e^{-(x-2t)^2} \right].$$

Exercise. *Solve the Cauchy problem*

$$\begin{aligned}u_{tt} &= c^2 u_{xx}, \quad -\infty < x < \infty, \quad t > 0, \\u(x, 0) &= a e^{-x^2} \sin x, \quad u_t(x, 0) = b \sin x, \quad -\infty < x < \infty.\end{aligned}$$

Solution. *We calculate*

$$\int_{x-ct}^{x+ct} \sin s ds = \cos(x-ct) - \cos(x+ct).$$

Therefore, D'Alembert's Solution (20) takes the form

$$u(x, t) = \frac{a}{2} \left[e^{-(x+ct)^2} \sin(x+ct) + e^{-(x-ct)^2} \sin(x-ct) \right] + \frac{b}{2c} (\cos(x-ct) - \cos(x+ct)).$$

Solution to the diffusion equation using the method of separation of variables

The Dirichlet Problem for the Diffusion Equation with homogeneous boundary conditions is given by

$$u_t = ku_{xx}, \quad 0 < x < L, t > 0, \quad (21)$$

$$u(0, t) = 0 \quad u(L, t) = 0, \quad t > 0, \quad (22)$$

$$u(x, 0) = f(x), \quad 0 < x < L, \quad (23)$$

We assume separable solution of (21) in the form

$$u(x, t) = X(x)T(t). \quad (24)$$

Putting u of (24) in the differential equation (21) we get

$$XT_t = kX_{xx}T,$$

and dividing by XT we have

$$\frac{T_t}{kT} = \frac{X_{xx}}{X}. \quad (25)$$

Note that, in equation (25), the left side is function of t only whereas the right side is function of x only. Therefore the equality in the equation (25) means that each side must be equal to a constant λ , say. Thus, we obtain following differential equations

$$\frac{dT(t)}{dt} = \lambda kT, \quad (26)$$

$$\frac{d^2X}{dx^2} = \lambda X. \quad (27)$$

The solution of (26) is

$$T(t) = Ae^{\lambda kt}, \quad (28)$$

A being an arbitrary constant.

Since we are looking for a bounded solution and $k > 0$, we must take $\lambda \leq 0$. Therefore

we can choose $\lambda = -\alpha^2$, α being real. Thus (28) takes the form

$$T(t) = Ae^{-\alpha^2 kt}, \quad (29)$$

and the equation (27) reduces

$$\frac{d^2 X}{dx^2} = -\alpha^2 X. \quad (30)$$

Solving the differential equation (30), we get

$$X(x) = B \cos(\alpha x) + C \sin(\alpha x), \quad (31)$$

where B and C are arbitrary constants. From the boundary conditions (22), we know that

$$u(0, t) = X(0)T(t) = 0 \Rightarrow X(0) = 0 \quad (32)$$

$$u(L, t) = X(L)T(t) = 0 \Rightarrow X(L) = 0. \quad (33)$$

In view of (31)-(33), we get $B = 0$, and

$$C \sin(\alpha L) = 0. \quad (34)$$

Consequently, a non-trivial solution (that is a solution for which $X(x) \neq 0$) for $X(x)$ can be found if and only if

$$\alpha = \alpha_n \equiv \left(\frac{n\pi}{L}\right) \quad \text{for } n = 1, 2, 3, \dots \quad (35)$$

for which we find

$$X(x) = X_n(x) \equiv C_n \sin\left(\frac{n\pi x}{L}\right) \quad \text{for } n = 1, 2, 3, \dots, \quad (36)$$

or some constant multiple of it. These special values of λ are called **eigenvalues** and the associated functions, $X_n(x)$, are known as **eigenfunctions**.

Multiplying the solution for $X_n(x)$ and $T(t)$ together finally yields a solution of (21),

$$u(x, t) = u_n(x, t) \equiv C_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi}{L}\right)^2 kt} \quad \text{for } n = 1, 2, 3, \dots \quad (37)$$

Since the equation (21) is linear, superposing all such solutions as (37) we may take the general solution of (21) as

$$u(x, t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi}{L}\right)^2 kt}. \quad (38)$$

Now the initial condition (23) implies

$$f(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right), \quad (39)$$

and

$$C_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (40)$$

Hence u in (38) together with (40) represents the solution of the IBVP (21)-(23).

Uniqueness of the solution to the diffusion equation

Theorem. *The solution $u(x, t)$ of the IBVP (21)-(23) is unique.*

Let $u_1(x, t)$ and $u_2(x, t)$ be two solutions of the IBVP (21)-(23). Then $U = u_1 - u_2$ satisfies the IBVP

$$\begin{aligned} U_t &= kU_{xx} & 0 < x < L, t > 0, \\ U(0, t) &= 0 \quad U(L, t) = 0 & t > 0, \\ U(x, 0) &= 0 & 0 < x < L. \end{aligned}$$

Let us define a function

$$W(t) = \frac{1}{2} \int_0^L U^2(x, t) dx. \quad (41)$$

Note that $W \geq 0$ with $W = 0$ only for the trivial solution $U(x, t) = 0$.

On differentiating W with respect to t , we get

$$\begin{aligned}\frac{dW}{dt} &= \int_0^L UU_t dx, \\ &= k \int_0^L UU_{xx} dx, \\ &= - \int_0^L (U_x)^2 dx + UU_{xx} \Big|_{x=0}^{x=L}.\end{aligned}$$

Since $U(0, t) = U(L, t) = 0$, we find that

$$\frac{dW}{dt} = - \int_0^L (U_x)^2 dx \leq 0$$

Now, we can conclude that W is decreasing function of t . From the condition $U(x, 0) = 0$, we have $W(0) = 0$. Therefore $W(t) \leq 0$ for all $t > 0$. But $W(t)$, by definition, is non-negative function. Therefore

$$W(t) \equiv 0 \quad \forall t > 0 \implies U \equiv 0 \quad \text{in } 0 \leq x \leq L, \quad t \geq 0.$$

Thus $u_1 \equiv u_2$. Hence the solution is unique.

Exercise. Solve

$$\begin{aligned}u_t &= u_{xx}, & 0 < x < \pi, t > 0, \\ u(0, t) &= 0 \quad u(\pi, t) = 0, & t > 0, \\ u(x, 0) &= \sin(3x), & 0 < x < \pi.\end{aligned}$$

Solution. In this case $k = 1$, $L = \pi$ and

$$\sin(3x) = \sum_{n=1}^{\infty} C_n \sin(nx).$$

Comparing the coefficients, we get $C_3 = 1$ and $C_n = 0$ when $n \neq 3$. Thus the solution is

$$u(x, t) = \sin(3x)e^{-9t}.$$

Exercise. *Solve*

$$\begin{aligned}u_t &= ku_{xx}, & 0 < x < \pi, t > 0, \\u(0, t) &= 0 \quad u(L, t) = 0, & t > 0, \\u(x, 0) &= Lx - x^2, & 0 < x < L.\end{aligned}$$

Solution. *In this case*

$$\begin{aligned}C_n &= \frac{2}{L} \int_0^L (Lx - x^2) \sin\left(\frac{n\pi x}{L}\right) dx. \\&= \frac{2}{L} \left(-\frac{2L^3}{n^3\pi^3} \cos(n\pi) + \frac{2L^3}{n^3\pi^3} \right) \\&= \frac{4L^2}{n^3\pi^3} (1 - (-1)^n).\end{aligned}$$

Hence the solution is

$$u(x, t) = \frac{4L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{(1 - (-1)^n)}{n^3} \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi}{L}\right)^2 kt}.$$